2024-25 First Semester
MATH1053 Linear Algebra I

Assignment 4

Due Date: Due on 15/Oct/2024 (Tuesday), at the end of lecture.

- Write down your **CHN name** and **student number**. Write neatly on **A4-sized** paper (*staple if necessary*) and **show your steps**.
- **Late submissions or answers without steps won't be graded.**

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1. Let
$$A = \begin{bmatrix} 3 & 4 \\ 2 & 6 \end{bmatrix}$$
   (a) Express A^{-1} as a product of elementary matrices.
   (b) Express A as a product of elementary matrices.

2. Use partitioned matrices to prove by induction that the product of two lower triangular matrices is also lower triangular. [Hint: A $(k+1) \times (k+1)$ matrix A_1 can be written in the form below, where a is a scalar, $\mathbf{v}$ is a $k \times 1$ real-valued vector, and A is a $k \times k$ lower triangular matrix.]
$$A_1 = \begin{bmatrix} a & \mathbf{0}^T \\ \mathbf{v} & A \end{bmatrix}$$

3. Compute the determinant of the following matrices using cofactor expansion,
$$A = \begin{pmatrix} -1 & -1 & 2 & 0 \\ 1 & 1 & 0 & -1 \\ 0 & 1 & -1 & 1 \\ 1 & 0 & -1 & 1 \end{pmatrix}, \; B = \begin{pmatrix} 2 & 1 & -3 \\ 4 & 0 & 1 \\ -2 & -1 & 4 \end{pmatrix}, \; C = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 4 & 8 \\ 2 & 4 & 8 \end{pmatrix}, \; D = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 1 & 6 & 4 & 8 \\ 1 & 3 & 0 & 0 \\ 2 & 6 & 4 & 8 \end{pmatrix}.$$

4. Find all possible values of λ so that the determinant $\left| \begin{pmatrix} 5 & 6 \\ 3 & -2 \end{pmatrix} - \lambda I_2 \right| = 0$.

5. Show that if A has two identical rows or two identical columns, then $\det(A) = 0$

6. (*Bonus!*) If $\mathbf{x}$ and $\mathbf{y}$ are nonzero vectors in $\mathbb{R}^n$ and $A = \mathbf{x}\mathbf{y}^T$, show that the row echelon form of A will have exactly one nonzero row.